varquanMS: local-scale, adaptive, variable-quantization-length compression method for radio interferometry data

Methods and validation manuscript. The earlier development manuscript is preserved as work_note_20261025_init_work.md. This version describes the implemented calibrated-data profile. Numerical results come from three verified fixed-epsilon archives and six CLEAN runs. Sizes are decimal MB.

Abstract

We present varquanMS, an experimental archive for radio-interferometric Measurement Sets using local scaling and adaptive integer precision. Baseline–frequency blocks within one correlation product store a complex median, a shared range, a power spacing parameter and a selected component width. A dimensionless epsilon-D criterion bounds reconstruction RMS relative to raw component median deviation, with maximum-error, bias and baseline guards. The precision ladder is 3, 4, 5, 6, 8, 10, 12, 14 and 16 bits per component plus exact float32, identified by a four-bit tag. Original UVW, ordering and retained metadata are reconstructed without coordinate regeneration or averaging. On a calibrated 15-integration OVRO-LWA observation, epsilon-D values 0.01, 0.018 and 0.04 produce visibility payloads of 22.859, 17.947, 17.433% of the original complex64 bytes and complete archives of 427.429, 357.127, 350.112 MB, respectively. The larger restored-image difference across two multiscale CLEAN runs is 1.260, 1.254, 1.151% of reference residual RMS, measured above 10° elevation. The calibrated profile explicitly discards raw/model columns and flagged values while retaining flags; its weights are exact on this dataset. These results measure agreement with the supplied calibrated reference, not sky truth or a general thermal-noise guarantee.

1. Introduction

An interferometer records complex correlations for many antenna pairs, frequency channels and times. The visibility array grows approximately as $N_{\mathrm{ant}}^2 N_\nu N_t N_p$, making storage and repeated data transfer costly. Complex64 uses two float32 components: eight bytes per visibility. Averaging can reduce volume, but also changes time or frequency resolution. We investigate lossy precision reduction while retaining the original sampling and row identities. The intended output is an independently decodable archive from which a new MS can be materialized for existing imaging software.

A Measurement Set includes coordinates, array-valued columns, descriptors, keywords and references to subtables. Correct archival reconstruction must account for this structure. Casacore also separates logical columns from physical storage managers, so repeated logical values do not by themselves establish wasted disk space. We report allocated payload and complete-file bytes separately. MS v2 specification, Casacore table system.

Normalization, nonlinear quantization and dithering are established techniques in visibility compression. Dysco combines these methods, including antenna/frequency normalizations, and evaluates their effects on LOFAR and MWA observations. Its published results motivate direct image-domain validation, but do not predict fidelity for the calibrated observation studied here. The present work investigates local precision selection, explicit exceptions, a dimensionless error rule and MS reconstruction. It does not claim normalization or nonlinear quantization as new, and a matched Dysco comparison has not yet been performed. Offringa (2016).

Array-based schemas such as XRADIO/MS v4 provide a useful separation between scientific coordinates and storage representation; this prototype exports MS v2 and does not claim MS v4 conformance. XRADIO overview. Blockwise low-bit formats and scale compression also appear in machine-learning quantization, including QLoRA. Here, all scale factors remain float32, no learned model is required, and model-quality results provide no scientific-fidelity assurance for visibilities. Dettmers et al. (2023).

2. Method

2.1 Inspection, partitioning and grouping

The input is inspected before encoding. The implemented regular path requires complex64 visibility cells with channels inside each cell and a homogeneous configuration within each partition. Multiple contiguous integrations are handled as separate partitions. Unsupported layouts fail explicitly. Ordered antenna pairs and distinguishing MS row context are preserved; antennas are not swapped and visibilities are not conjugated to canonicalize a baseline.

The internal visibility axes are baseline/row, channel and correlation. Product names and order are read from POLARIZATION metadata. Autocorrelation and cross-correlation rows are separated. For each product p, a stable amplitude ordering uses

$$A_{b,p} = \mathrm{median}_{\nu \in \mathcal{V}_{b,p}} \mid V_{b,\nu,p} \mid,$$

where $\mathcal{V}$ excludes flagged and nonfinite samples. Original row index breaks ties, and rows with no eligible values form a deterministic special group. Each baseline spectrum stays intact. The stored permutation obeys

$$V^{\mathrm{archive}}_{j,\nu,p} = V^{\mathrm{source}}_{\pi_p(j),\nu,p}, \qquad V^{\mathrm{restored}}_{\pi_p(j),\nu,p} = \hat{V}^{\mathrm{archive}}_{j,\nu,p}.$$

A quantization block contains at most B consecutive archived baselines and F consecutive channels for one product. Exact edge bounds and class boundaries are stored. The present experiment fixes $B = 128$ and $F = 48$, independently per product. No time/channel averaging is performed. Sorting changes which values share factors; the permutation itself removes no information and has a measured byte cost. The implementation also supports original, baseline-length, orientation, antenna and amplitude-bucket orders, but this three-tolerance experiment does not compare those alternatives.

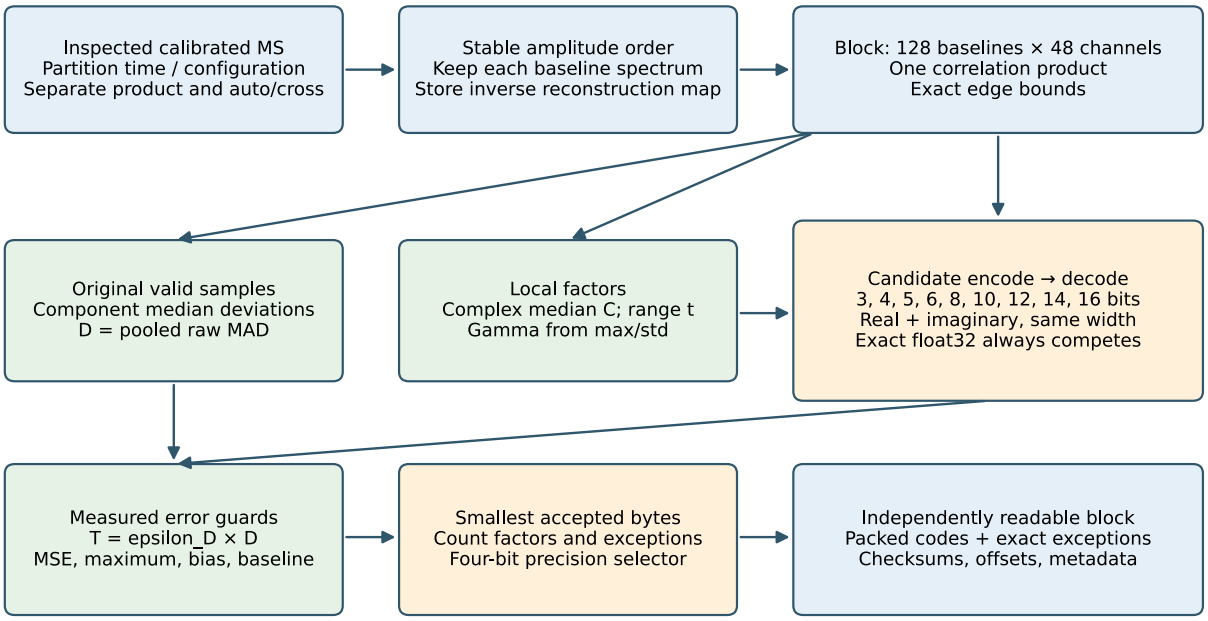


Figure 1. Spectra are ordered within one configuration, integration, product and auto/cross class. Each block is independently encoded, reconstructed and checked; the smallest accepted serialized candidate is stored with its reconstruction map.

2.2 Local centre, range and power spacing

For block q, store two component medians $C_q = C_{q,R} + iC_{q,I}$, a shared range $t_q \geq 0$ and a power parameter γ_q. The encoder computes the medians from eligible values, excluding flags, nonfinite values and protected signed-zero samples, then rounds the medians to float32. Residuals are calculated against these stored medians. Two range candidates are tested: the maximum and the 99.5th percentile of $\max(|\,\Re(V - C_q)\,|, |\,\Im(V - C_q)\,|)$. Each range is rounded upward to float32. Values outside a candidate range are preserved through exact exceptions.

For either real component x, define $y = (x - C)/t_q$ on $[-1, 1]$. Let

$$M_b = 2^b - 1, \qquad u_k = \frac{2k - M_b}{M_b}, \qquad k \in \{0, ..., M_b\}.$$

The reconstruction levels are

$$\hat{x}_k = C + t_q \,\mathrm{sgn}\,(u_k)\;|\;u_k\;|^{1/\gamma_q}\;.$$

All 2^b unsigned codes are used. The residual grid is symmetric and spans $[-t_q, t_q]$, without a separate sign bit or a duplicate signed-zero code. Since M_b is odd, the grid has no exact residual-zero level for $t_q > 0$. This is a midrise grid: an ordinary value at the centre can acquire a small error. Negative zeros are protected exactly through exceptions. If $t_q = 0$, the decoder copies the stored component centres directly. Entirely zero or otherwise compressible blocks may instead select the exact representation.

For $\gamma_q = 1$, the reconstruction spacing is uniform, $\Delta_b = 2t_q/(2^b - 1)$. For $0 < \gamma_q < 1$, levels concentrate near the centre:

$$\left|\frac{d\hat{x}}{du}\right| = \frac{t_q}{\gamma_q}\;|\;u\;|^{1/\gamma_q - 1}\;.$$

The candidate encoder first computes the continuous coordinate

$$k^* = \frac{M_b}{2}[1 + \operatorname{sgn}(y) \mid y \mid^{\gamma_q}].$$

It reconstructs the adjacent integer codes $\lfloor k^* \rfloor$ and $\lceil k^* \rceil$ using the stored factors and chooses the closer reconstructed float32 value. Equal errors select the even code. Choosing by reconstructed values matters for nonlinear spacing; ordinary rounding of k^* alone need not choose the closest level in data units. Unflagged outliers are never silently saturated. Endpoint clipping only protects the integer lookup from floating-point roundoff after the range/exception decision.

The tested gamma rule uses the maximum absolute component residual divided by pooled within-component population standard deviation, r_q. Both statistics use all eligible residuals before selecting range exceptions, with float64 arithmetic and variance normalization `ddof=0`. It selects

$$\gamma_q = \begin{cases} 1, & r_q \le 4 \text{ or the ratio is undefined,} \\ 1/2, & 4 < r_q \le 8, \\ 1/4, & r_q > 8. \end{cases}$$

This is a deterministic encoder heuristic, not a noise model. The selected gamma is stored, so decoding never estimates it again. Full gamma search is available as a separate experiment. The decoder evaluates normalized codes and powers in float64, using explicit squares for gamma $1/2$ and fourth powers for gamma $1/4$, then multiplies by the stored range, adds the stored centre and rounds once to float32. The reference implementation fixes the operation order.

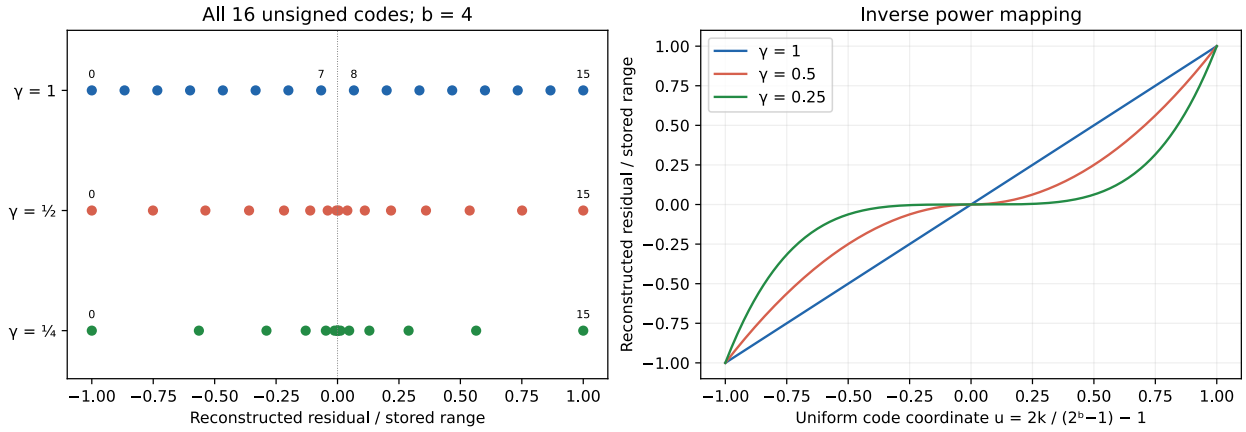


Figure 2. Illustrative four-bit reconstruction levels after median subtraction. Every row contains 16 distinct unsigned codes. Smaller gamma increases resolution near the centre while increasing spacing near the range endpoints. The experiment also allows eight other component widths.

2.3 Choosing precision with epsilon D

The reference statistic for acceptance is computed from the original finite, unflagged samples, before outlier selection. Define raw component deviations

$$d_{q,R} = \operatorname{median} \mid \Re V - \operatorname{median}(\Re V) \mid, \qquad d_{q,I} = \operatorname{median} \mid \Im V - \operatorname{median}(\Im V) \mid,$$

$$D_q = \sqrt{\frac{d_{q,R}^2 + d_{q,I}^2}{2}}, \qquad T_q = \epsilon_D D_q.$$

These medians use float64 arithmetic and no Gaussian conversion factor. D_q describes signal variation; it is not a thermal-noise estimate. It is separate from the rounded median used

for coding. Multiplying all visibilities by a real amplitude factor scales D_q and the permitted absolute error by its magnitude, subject to finite-precision effects. Thus the tolerance has no fixed amplitude unit and requires no denominator floor near an individual zero visibility.

For reconstructed error $e = \hat{V} - V$, the tests are

$$\frac{1}{2N_q}\sum_{n\in\mathcal{V}_q} |\, e_n \,|^2 \le T_q^2, \qquad \max_{n\in\mathcal{V}_q} |\, e_n \,| \le 10T_q,$$

$$\left|\frac{1}{N_q}\sum_{n\in\mathcal{V}_q} e_n\right| \le T_q, \qquad \max_{b\in q}\frac{1}{2N_{q,b}}\sum_{n\in\mathcal{V}_{q,b}} |\, e_n \,|^2 \le 2T_q^2.$$

Only baselines with eligible samples contribute to the last expression. These are pooled real/imaginary constraints; separate component errors are measured but there is no independent per-component relative ceiling. A zero D_q imposes zero measured error, even when a sparse nonconstant block has zero MAD. Empty valid sets have zero measured errors. Exact fallback is always available.

The precision ladder is

$$b \in \{3, 4, 5, 6, 8, 10, 12, 14, 16\} \cup \{\text{exact float32}\}.$$

Each bit width and range candidate is actually reconstructed and tested. The encoder chooses the candidate with the fewest serialized bytes, including the header, codes, exception stream and optional lossless compression. Exact storage also competes on size and wins equal-size ties. A larger epsilon generally permits coarser precision, but bytes and image error are measured rather than inferred from the nominal width. Precision is common to the real and imaginary components within a block. A b-bit component codec uses $2b$ payload bits per complex sample before exceptions and other overhead.

The general interface also supports sampled global std, block std and uncentered block RMS references through ϵ_G, ϵ_S and ϵ_R. Those coefficients are disabled in the present experiment. Their sampling and combination study is documented in the 16% study; no fixed absolute-error threshold is active here.

2.4 Data format and precision tags

The scientific schema is independent of its HDF5 prototype container. The archive holds typed table descriptors and keywords, exact supported subtable values, original coordinates, row/cell maps, permutations, flags, weights, visibility streams, error summaries and integrity inventories. Shared subtable content is stored once when compacting the integration partitions. Explicit little-endian dtypes and JSON metadata avoid executable object serialization.

Each block has a 16-byte base header containing a precision selector, compression identifier, mapping identifier, complex sample count, exception count and a float32 range or step. Power blocks append two float32 centre components and float32 gamma, making 28 bytes before the body. An exact block uses the 16-byte base header and original complex64 bits. The first byte of the new header is `0xA0 | tag`, with the lower four bits interpreted as follows:

Tag	Interpretation
0	Exact complex64
1–9	3, 4, 5, 6, 8, 10, 12, 14, 16 bits/component, respectively
A	Legacy two-bit representation, outside the tested ladder
B–E	Reserved; decoder rejects
F	Invalid/error; never a successful encoded block

Four bits suffice to select the precision, not to represent every visibility. The upper nibble identifies this header version; aligned header sizes remain 16/28 bytes. The format does

not claim an additional half-byte saving per block. Historical literal-width headers remain decodable, while unsupported tags fail explicitly.

Codes are interleaved real then imaginary and packed least-significant bit first. Each width occupies its actual number of bits; 14-bit codes can cross three bytes. Unused final high bits must be zero. Exact exceptions store increasing uint32 sample positions and original eight-byte complex64 values. They restore protected special-value bits and outliers after ordinary reconstruction. Block bodies use DEFLATE only when it reduces their size. Offset indexes and per-block CRC32 allow independent reads; dataset SHA-256, container checksums and a manifest inventory provide further corruption detection. These checks detect accidental damage; they are not authentication signatures. The decoder needs no remote service, learned model or downloaded codebook. A standard-library reference block decoder and small literal conformance fixtures accompany the implementation.

2.5 Compression, decompression and calibrated-only policy

Compression proceeds through inspection, context partitioning, grouping, candidate selection, payload writing and source-aware verification. Partial outputs remain distinguishable from completed archives. The source MS is never overwritten. Decompression checks inventories and block bounds, reconstructs values, applies exact exceptions and inverse permutations, rebuilds subtable references, and writes a new MS. Original UVW float64 values are retained in metres; no antenna position, time or frequency calculation regenerates them. Bitwise-identical coordinates may be deduplicated, without tolerance-based merging.

The calibrated-only profile explicitly retains CORRECTED_DATA and removes the original DATA and MODEL_DATA. Original flagged visibility values are discarded and represented by positive-zero placeholders, while FLAG, FLAG_ROW and FLAG_CATEGORY remain preserved. It retains sample presence, array shapes and resolution. Consequently this profile is not an exact round trip of the original MS, even when the retained visibilities choose exact storage. On export, DATA is recreated from decoded CORRECTED_DATA for MS compatibility; original raw DATA is not recovered. Remaining supported metadata and auxiliary columns stay exact.

Weights use an independent four-bit codebook with checked float32 fallback; weight approximation, if needed, follows its separately recorded tolerance. The weights of this particular observation are reconstructed bitwise exactly. No quantization error estimate overwrites original weights. Optional removal of flags is a distinct destructive profile and is not enabled in these experiments.

For the supported multi-integration case, the reference workflow is:

```
export PYTHONPATH=src
python scripts/timeslot_archive.py encode INPUT.ms blocks.bundle \
  --config configs/calibrated-epsd-0018.json --report encode.json
python scripts/timeslot_archive.py compact blocks.bundle archive.h5 \
  --report compact.json
python scripts/timeslot_archive.py decode archive.h5 RESTORED.ms \
  --report decode.json
```

All outputs must be new paths. Decoding receives the archive alone. Source-aware validation is a separate step and does not supply missing reconstruction data.

3. Validation on calibrated OVRO-LWA data

3.1 Observation and experiment design

The inspected input `lwa-15slot-flat.ms` contains 15 contiguous integrations, 931,920 rows and 352 antennas. Each integration has 62,128 ordered antenna pairs: 352 autocorrelations and 61,776 cross-correlations. Each cell has 48 channels and four products in metadata order XX, XY, YX, YY. Channel centres range from 63.965576 to 68.463623 MHz at 95.703125 kHz

spacing. Original TIME and TIME_CENTROID values and their UTC references are retained. The table contains 926,641 bitwise-distinct original UVW vectors.

CORRECTED_DATA contains 178,928,640 complex samples, or 1,431,429,120 logical bytes. There are 41,752,612 flagged samples. The complete source occupies 4,428,203,949 bytes, including raw and model columns that the calibrated profile deliberately omits. These different denominators must not be conflated. Source HISTORY records CASA `applycal`; this establishes the supplied calibration history, not an independent validation of the calibration or flux scale. Input inspection.

We fix all grouping, median, gamma, exception and preservation settings and test $\epsilon_D = 0.01$, 0.018 and 0.04 with the expanded nine-width ladder. The coefficients are specified in advance and are not retuned to a byte target or selected from image outcomes. Each case reads the original calibrated source; none recompresses a previous quantized result. All use 128×48×1 blocks, independent amplitude ordering, the max/std gamma rule, range quantiles 1 and 0.995, and exact fallback. Encoder workers are set to 24. Exact configuration files accompany the results.

3.2 Reconstruction and imaging checks

The validation suite exercises every width, partial blocks, special values, flags, signed zeros, exact fallback, malformed/truncated streams and both header representations. Independent standard-library, portable NumPy and compiled reconstructions are compared. Native modules built before a requested width fall back to portable encoding. The current full suite passes 114 tests. Source-aware archive/export checks compare original supported metadata, coordinates, flags and weights exactly, and independently recompute normalized visibility error criteria. Source checksums are checked against the original import inventory.

Imaging uses WSClean 3.7 in the same recorded container for reference and quantized MSs. A private MS copy protects each input from imaging side effects. The uncompressed reference and every reconstructed case use Stokes I, 2048×2048 pixels, four-arcminute pixels, Briggs robust 0 and multiscale CLEAN. The remaining settings are eight threads, scale bias 0.8, 150,000 iterations, mgain 0.95, auto-threshold 0.5, auto-mask 3, local RMS, a 10° horizon mask, 40% memory setting, reordering and no model-column update. Candidate images use the reference restoring beam. Auto-masks are estimated independently, so deconvolution-path changes contribute to the measured differences. WSClean is an established wide-field imager; these particular validation settings and results belong to this experiment. Offringa et al. (2014).

The comparison mask requires elevation above 10° for all 15 integrations and contains 2,245,726 pixels. Let I_0 be the restored reference image, I_ϵ a quantized restored image, and R_0 the reference residual map. We report the uncentered, unclipped statistics

$$E_{\mathrm{image}} = \mathrm{RMS}_{\mathcal{M}}(I_\epsilon - I_0), \qquad \eta_{\mathrm{residual}} = 100\frac{E_{\mathrm{image}}}{\mathrm{RMS}_{\mathcal{M}}(R_0)}.$$

The reference residual RMS is 1.031244811 Jy/beam; the restored-reference RMS is 1.898403101 Jy/beam. Residual-normalized difference is not fractional flux loss, per-pixel relative error, or a thermal-noise budget. The residual contains sky and deconvolution structure. We additionally compare dirty images, faint-region RMS, reference-peak changes and visibility errors by baseline, product and frequency. The reference defines faint pixels below 1% of its peak absolute intensity. Each compressed MS is CLEANed twice to record observed repeat variation; two runs do not establish statistical confidence intervals.

3.3 Measured size, precision and image differences

εD	Payload MB (% of complex64)	Complete archive MB	Difference mJy/beam, first / repeat	Difference / reference residual RMS %, first / repeat
0.01	327.207 (22.859%)	427.429	12.9866 / 12.9944	1.2593 / 1.2601
0.018	256.906 (17.947%)	357.127	9.0301 / 12.9284	0.8756 / 1.2537
0.04	249.542 (17.433%)	350.112	11.8731 / 11.8666	1.1513 / 1.1507

εD	Dirty difference mJy/beam	Worst reference-peak change %	Worst faint-region RMS difference %	Candidate residual RMS Jy/beam, first / repeat
0.01	1.5080	0.00020	1.0430	1.030953 / 1.030953
0.018	2.9380	0.00150	1.0392	1.031052 / 1.031412
0.04	5.4572	0.00244	0.9556	1.031218 / 1.031221

εD	Cross XX RMSE / signal RMS %	Cross XY %	Cross YX %	Cross YY %
0.01	0.59372	0.38169	0.48786	0.45558
0.018	0.77274	0.84894	0.82884	0.82922
0.04	2.35604	1.43210	1.87325	1.78090

Visibility ratios divide component reconstruction RMSE by the original component signal RMS, pooling all finite unflagged cross-correlations in each product. They are distinct from the local MAD-normalized acceptance criterion. Cross-hand visibility statistics do not constitute a polarized-image validation.

All three archives passed full-source export verification. Original supported metadata, UVW, flags and this observation’s weights passed exact checks; all 108 source-file hashes still match. All six CLEAN runs reached the final threshold, with three major iterations each. The new 5/10/14-bit choices encode 0, 105,035,520 and 0 complex samples, respectively, summed across the three separate experiments. Exact blocks often represent highly compressible zeros, so sample fractions and byte fractions differ. Complete distributions and auto/cross splits are retained in the machine-readable experiment report.

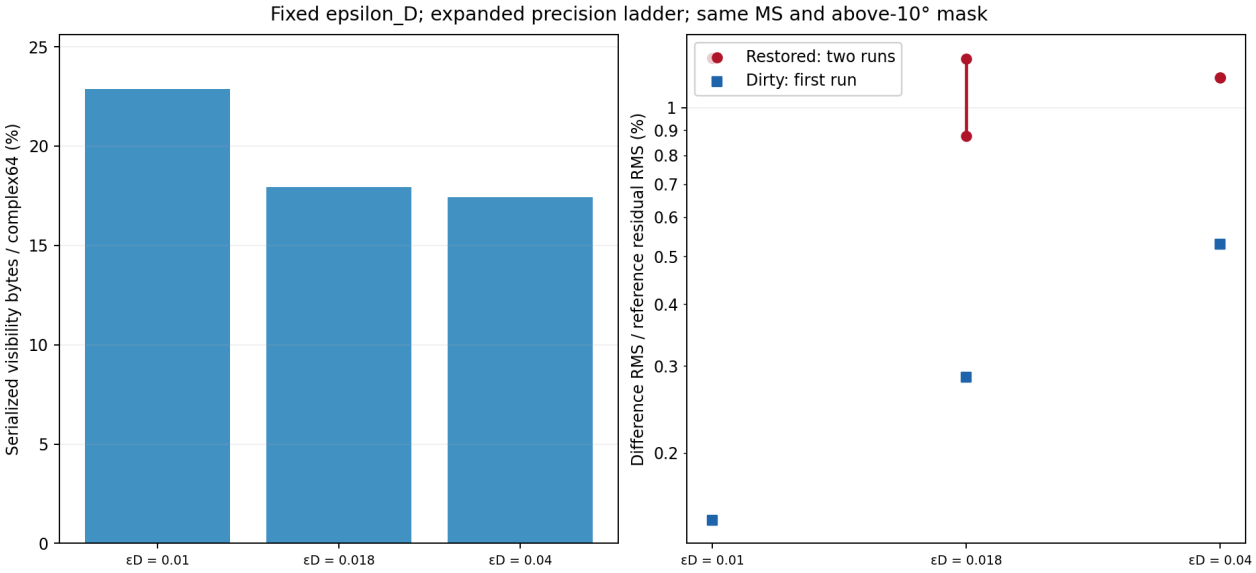


Figure 3. Fixed epsilon-D values produce measured sizes and image errors; there is no imposed byte target. Red points show two CLEAN runs. Blue squares show the first dirty-image difference, all normalized by the same reference residual RMS.

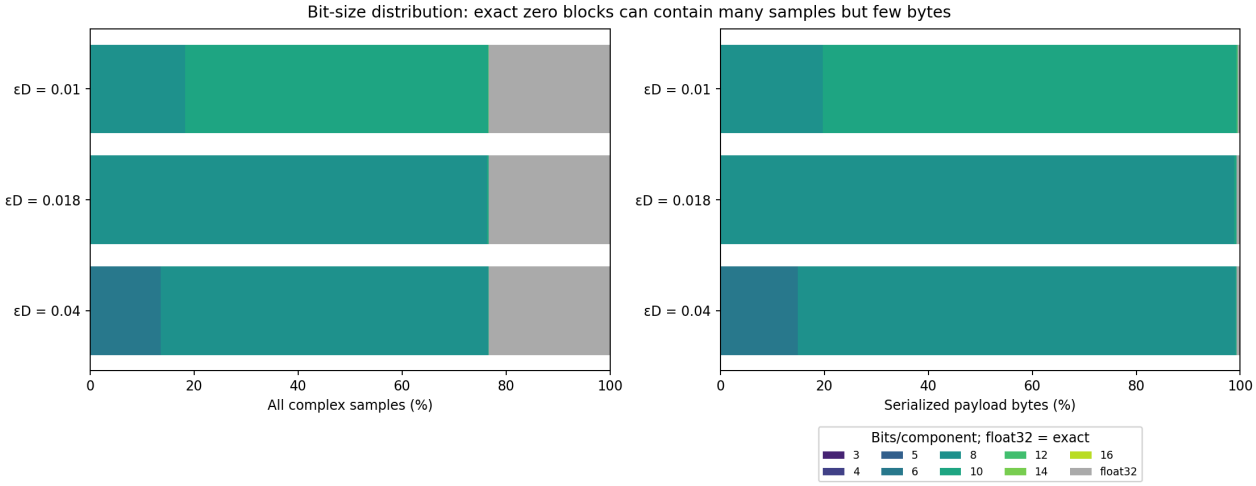


Figure 4. Selected component widths by number of complex samples and by serialized payload bytes. Float32 denotes exact storage, including cheaply compressed zero blocks. Metadata outside visibility blocks is accounted for separately.

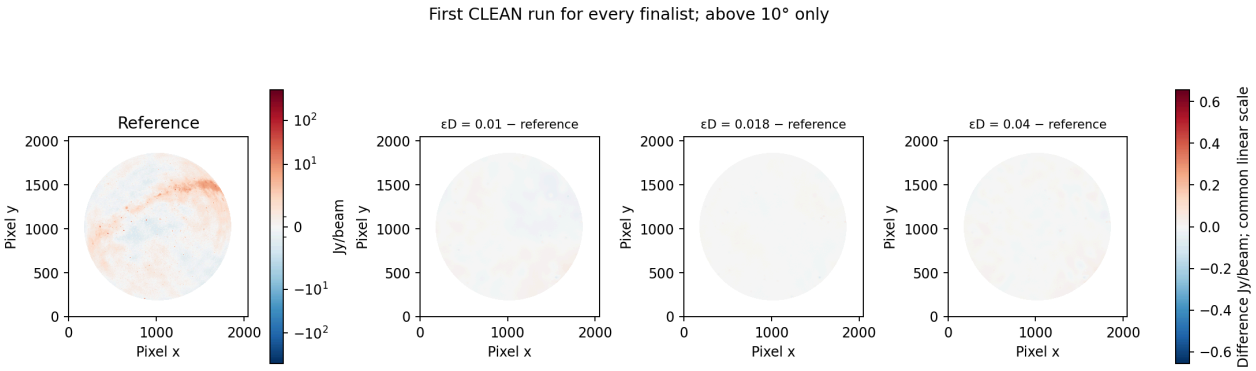


Figure 5. Reference and first-run quantized-minus-reference restored images above 10°. Difference panels share one linear colour scale; the reference uses its own symmetric logarithmic display scale. Pixels outside the mask are excluded.

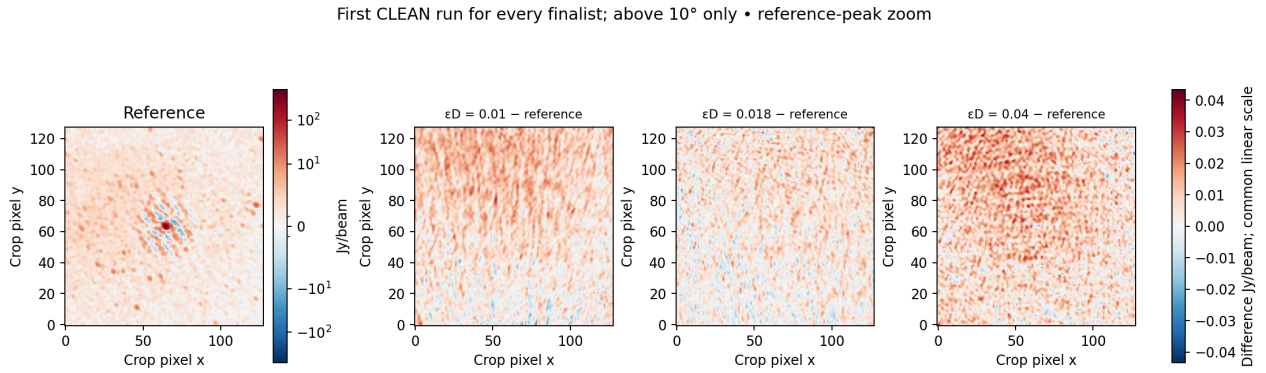


Figure 6. The same comparisons in a fixed 128×128-pixel crop around the reference peak. The reference peak defines the crop independently of compressed results.

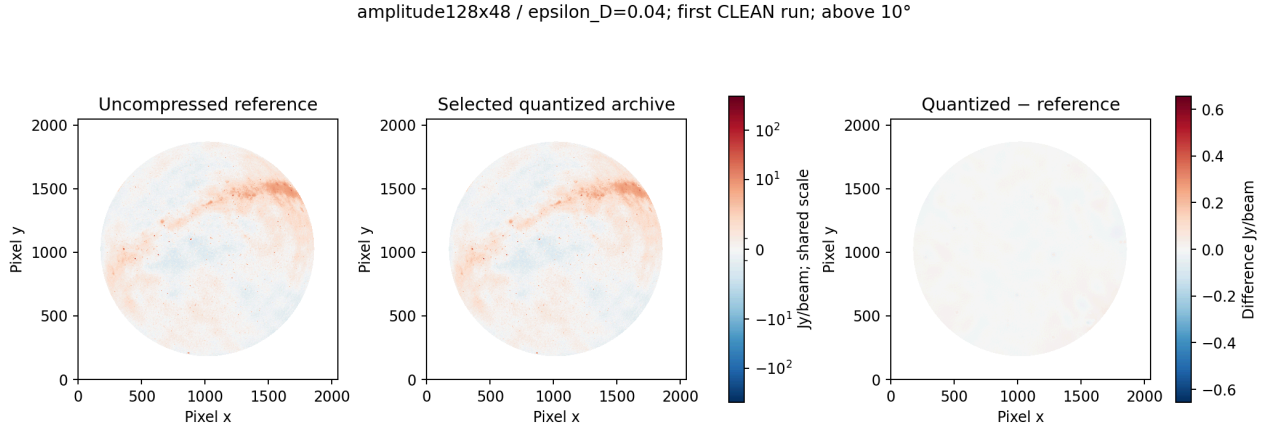


Figure 7. Reference, quantized image and difference for epsilon-D 0.04, the smallest worst-of-two global image-RMS difference in this three-point test. This selects a display case by error, without asserting an optimal size–error tradeoff.

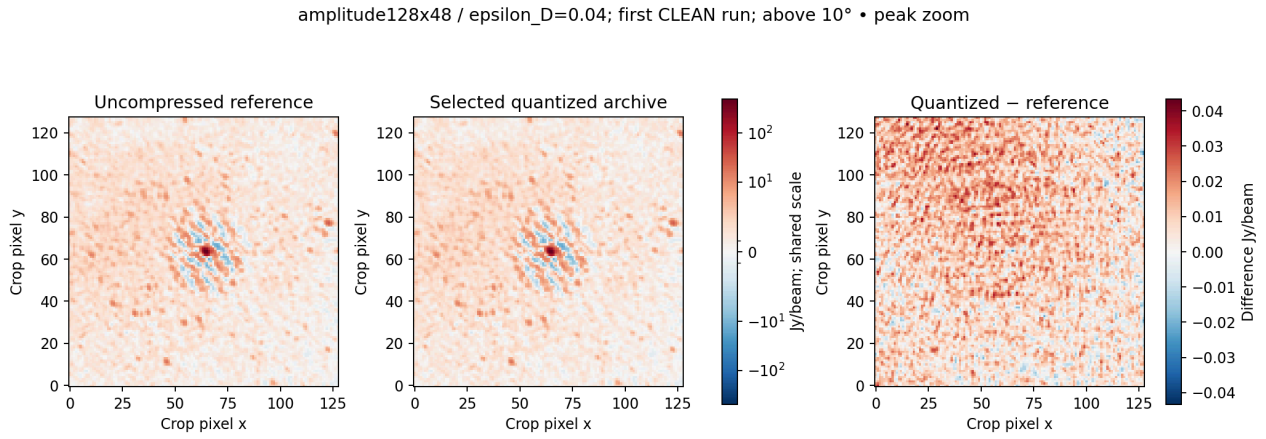


Figure 8. Reference-peak crop for the same case. The first two panels share an intensity scale; the difference has a separate, explicitly labelled linear scale.

3.4 Timing and byte accounting

εD	Sum partition encode s	Global sampling s	Verified bundle wall s	Compaction s	MS export s	CLEAN s, first / repeat
0.01	196.45	7.45	456.48	33.25	58.61	174.66 / 185.93
0.018	202.66	7.18	756.66	23.80	48.32	166.18 / 172.85
0.04	178.68	6.97	456.93	31.88	55.97	288.42 / 213.19

Timings cover all 15 integrations on a shared Intel Xeon Silver 4214 host with 48 logical CPUs and approximately 376 GiB RAM. Grouped encoders use 24 workers; CLEAN uses eight threads. Archive pipelines and imaging overlap, and caches are not evicted. These measurements describe the observed workflow rather than an isolated speed ranking. Verified-bundle wall time includes source copying, inspection, encoding, global-reference sampling and verification; compaction and MS export are additional. Global references are still recorded even though G is disabled. The full-conversion 15-second target remains unmet.

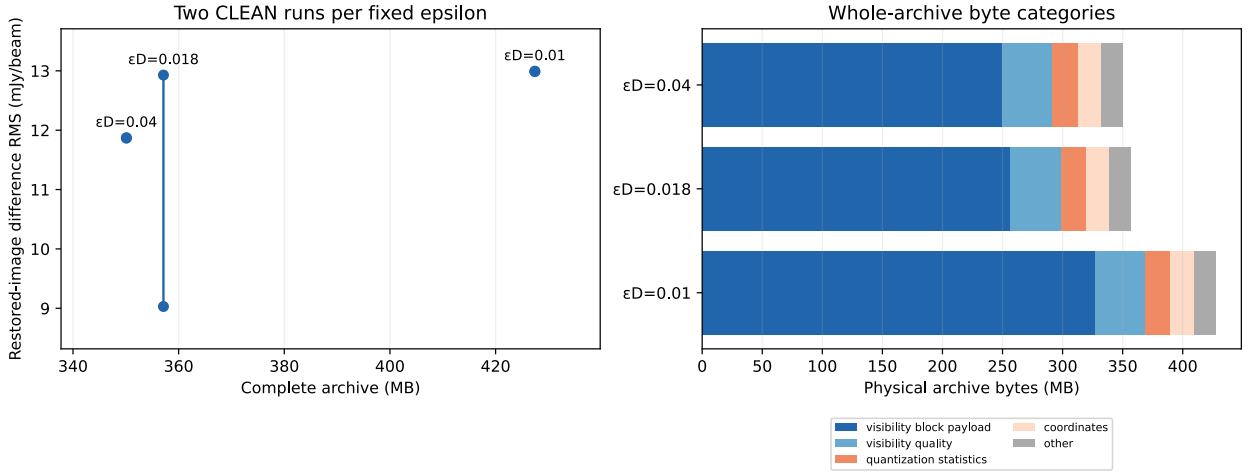


Figure 9. Complete-file size versus the two restored-image RMS differences, and physical byte categories. Factors and exceptions are inside visibility payload; quality records, coordinates, permutations and other metadata remain part of the complete archive. Summing only nominal component bits would understate storage.

4. Conclusion and discussion

Local median subtraction, full-range unsigned power levels and an explicit ϵ_D acceptance rule provide a compact and interpretable encoder policy. The coefficient expresses allowed error relative to block variation, while actual reconstruction checks determine which precision is acceptable. Nine quantized choices plus exact fallback fit within a four-bit precision selector. Broader precision coverage does not remove the need to count factors, exceptions, permutations, indexes, coordinates and error summaries in the archive size.

The three specified epsilon-D values yield complete archives between 350.112 and 427.429 MB. Their worst-of-two image differences range from 1.151% to 1.260% of the reference residual RMS on the fixed elevation mask. This is a measured three-point size–error comparison with constant grouping and codec settings. It does not isolate the effect of expanding the ladder from the earlier six-width experiments, whose tolerances also differ. No result here demonstrates a universal optimum or a scientific acceptance threshold. Preserved exact data, explicit profile losses and complete byte counts make the tradeoffs reviewable for this observation.

The epsilon-D 0.01 case has a smaller first-run dirty-image difference (0.146% versus 0.529% of reference residual RMS) than epsilon-D 0.04, yet its worst restored-image difference is larger (1.260% versus 1.151%). Both outcomes are retained: tighter visibility bounds do not impose a monotonic ordering on the CLEAN results. The repeated runs quantify observed variation but do not establish statistical significance or a causal model for this reversal.

A single block statistic cannot establish every future science requirement. Median deviation can vanish for sparse nonconstant data, forcing exact storage; strong and weak products can have different scientific tolerances. Deterministic rounding can create structured errors despite a small mean-square bound. Image agreement depends on visibility weighting and nonlinear deconvolution as well as visibility error, so dirty-image agreement alone is insufficient. These are reasons to retain direct image checks and explicit maximum/bias guards.

The present validation covers joint Stokes I on one supplied calibrated observation. It does not establish polarimetric leakage, narrow spectral-feature, calibration-solution or transient-light-curve fidelity, and it contains no held-out observation. CASA compatibility is limited to the checks documented in the repository; the new experiments directly exercise python-casacore reconstruction and WSClean. A matched Dysco benchmark, additional observations and isolated runtime measurements remain necessary before choosing general archival defaults. The conservative default remains lossless.

The public repository contains the format, reference decoder, configurations, conformance tests, machine-readable validation and figure-generation scripts. Detailed operational history is retained in development notes and the preserved initial manuscript, rather than being appended to this methods paper.